

Day - 3

9. Risk Factors vs Severity (Regression Analysis)

Chart Used:

Regression Plot (Scatter + Regression Line)

```
sns.regplot(x=x, y=y)
```

Why Regression Plot?

- Risk factors are continuous
- Severity score is continuous
- Want to examine linear relationship

Statistical Model:

Used:

```
linregress()
```

It gives:

- Slope
- Intercept
- R-value
- R^2
- p-value

What to Explain:

- **Slope** → Direction of impact
- **R^2** → Strength of relationship (0 to 1)
- Higher R^2 = stronger predictor

Business Insight:

Identifies strongest lifestyle/genetic predictor of cancer severity.

10. Proportion of Early-Stage Diagnosis by Cancer Type

Objective

To determine how often cancers are diagnosed in early stages (Stage 0 & Stage I).

What I Did

- Filtered dataset by cancer type
- Calculated:
 - Total cases
 - Count of Stage 0
 - Count of Stage I
- Computed proportion percentage

Code Hint

- Use filtering with `data[data["Cancer_Type"] == "..."]`
 - Use `value_counts()`
 - Use `.get("Stage 0", 0)` for safe extraction
 - Calculate percentage
-

Interpretation Logic

Why This Matters

Early-stage diagnosis:

- Increases survival probability
 - Reduces treatment cost
 - Improves healthcare efficiency
-

Business Meaning

If a cancer type shows:

- High early-stage proportion → Good screening system
- Low early-stage proportion → Late detection issue

This can guide:

- Public health campaigns
- Screening program investments
- Country-level healthcare improvements

11. Identifying Key Predictors of Cancer Severity & Survival

Objective

To determine which features influence:

- Survival Years
- Target Severity Score

What I Did

- Calculated Pearson correlation (linear relationship)
- Calculated Spearman correlation (monotonic relationship)
- Combined results for comparison

Code Hint

- Use `data[features + targets].corr(method="pearson")`
 - Repeat for `spearman`
 - Slice only target columns
 - Combine using `pd.concat()`
-

Interpretation of Your Output

Survival Years

Almost zero correlation with all features.

Example:

Age → -0.0001

Smoking → 0.0013

Meaning:

Survival years appear randomly distributed in this synthetic dataset.

In real-world healthcare data, this would not be expected.

Target Severity Score

Strong Positive Correlations:

Factor	Correlation
Smoking	0.48

Genetic Risk 0.47

Air Pollution 0.36

Alcohol Use 0.36

Obesity Level 0.25

Key Predictors (Ranked)

1. Smoking
2. Genetic Risk
3. Air Pollution
4. Alcohol Use
5. Obesity Level
6. Age (No impact)

Business Insight

Severity is largely driven by:

- Behavioral risk factors (Smoking, Alcohol)
- Genetic predisposition
- Environmental exposure

This supports:

- Preventive healthcare strategy
- Lifestyle modification campaigns
- Environmental regulation policies

TASK 1: Multivariate Severity Analysis (Not Just One Factor)

Objective:

Instead of analyzing one risk factor at a time, analyze **combined effect**.

Task:

Use these features:

```
['Age', 'Genetic_Risk', 'Air_Pollution', 'Alcohol_Use', 'Smoking', 'Obesity_Level']
```

- 1.
2. Build a **Multiple Linear Regression model**
3. Predict:
 - `Target_Severity_Score`

Deliverables:

- Model R^2 score
- Adjusted R^2
- Coefficient table
- Interpretation:
 - Which factor increases severity the most?
 - Which factor has minimal impact?

Business Question:

If you were a hospital administrator, which lifestyle factor would you target first?

TASK 2: Survival Prediction Analysis

Objective:

Find what impacts `Survival_Years` the most.

Task:

1. Build a regression model predicting:
 - `Survival_Years`
2. Compare:
 - Model predicting Severity
 - Model predicting Survival

Deliverables:

- Model Accuracy (R^2)
- Feature importance ranking
- Compare top 3 predictors for:
 - Severity
 - Survival

Business Thinking:

Are severity and survival influenced by the same factors?

TASK 3: Cancer Type Risk Profiling**Objective:**

Find which cancer type is most dangerous.

Task:

1. Calculate:
 - Average Severity Score per Cancer Type
 - Average Survival Years per Cancer Type
2. Rank Cancer Types:
 - Highest Severity
 - Lowest Survival

Visualization:

- Barplot (Severity by Cancer Type)
- Barplot (Survival by Cancer Type)

Business Insight:

Which cancer type should government screening programs prioritize?

TASK 4: Early Stage Detection Efficiency (All Cancer Types)

You only did Lung & Liver.

Task:

1. For every `Cancer_Type`:
 - Calculate % of early-stage (Stage 0 + Stage I)

Create a summary dataframe:

`Cancer_Type` | `Early_Stage_%`

- 2.
3. Sort descending

Business Insight:

Which cancer has best early detection system?
Which needs awareness programs?

TASK 5: Cost vs Survival Analysis

Objective:

Is higher treatment cost linked to better survival?

Task:

1. Correlation between:
 - `Treatment_Cost_USD`
 - `Survival_Years`
2. Plot regression
3. Interpret slope

Critical Thinking:

Does expensive treatment guarantee longer life?